

5. Action Selection

Receding - Horizon:

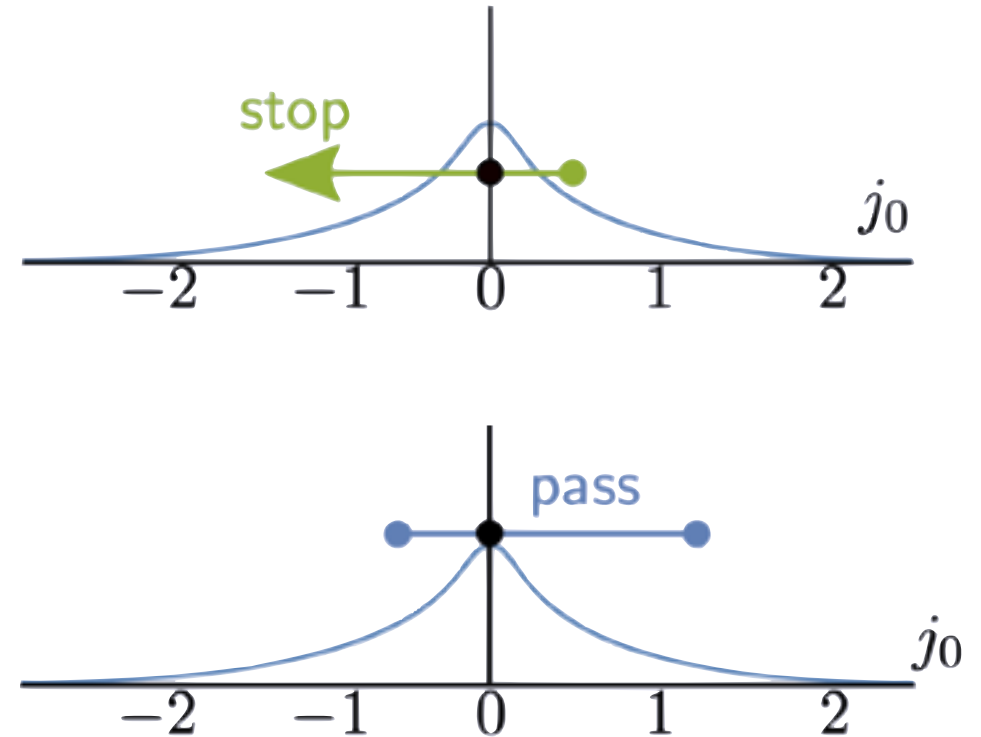
Motor plans are continuously updated $\rightarrow j(0)$ is the **instantaneous** control.

Human - Directed Jerk:

90% of observation lays in the interval of $\left[-1\frac{m}{s^3}, 1\frac{m}{s^3}\right]$, so we

choose to adopt the solution that presents the value of jerk closer (or belonging) to this interval.

In some cases we need to calculate some new **stop/pass** primitives with $j(0) = 0$.



6. Low Level Control

The agent follows the primitives using a **PI control**.

Define the error:

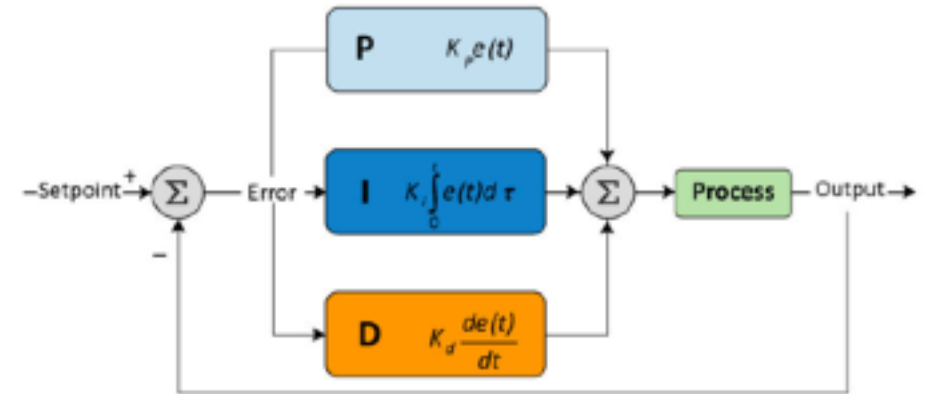
$$e = a_{req} - a;$$

Compute the integral of the error:

$$e_{int} = e_{int} + e \cdot dt$$

Requested pedal:

$$Req_{ped} = e \cdot P_{gain} + e_{int} \cdot I_{gain}$$



Wind - Up problem:

If the integral term accumulates a **large** error (i.e. when the vehicle stops at the traffic light), it can cause **instability** or slow convergence to the target (i.e. **not** moving at green light) → so, the integral error is **imposed** to be **zero** when a velocity lower than few m/s is reached.